MEASURES OF LOCATION

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MEASURE OF CENTRAL TENDENCY

The diagrammatic representation of a set of data can give us some impression about its distribution. Even then there remains a need for a single quantitative measure which could be used to indicate the center of the distribution . An average is a single value which represent a set of data or a distribution as a whole. It is more or less central value round which the observations in the set of data or distribution tend to cluster. Such a central value is also called a measure of central tendency or measure of location.

MEASURE OF CENTRAL TENDENCY

QUALITIES OF A GOOD AVERAGE

(i) Rigorously defined

(ii) Based on all the observations in the data

(iii) Simple to understand and easy to interpret

(iv) Not much influenced by abnormally large or small observations

MEASURE OF CENTRAL TENDENCY

TYPES OF AVERAGES

1. **Arithmetic mean**
2. **Geometric mean**
3. **Harmonic mean**
4. **Median**
5. **Mode**

MEASURE OF CENTRAL TENDENCY

<u>Arithmetic Mean</u>

It is obtained by dividing the sum of all the observations by the total number of observations. It is denoted by " $\overline{X}$ ".

For ungrouped data

$$\overline{X} = \frac{x_1 + x_2 + x_3 + \dots x_n}{n} = \frac{\sum X}{n}$$

For grouped data

$$\overline{X} = \frac{f_1 x_1 + f_2 x_2 + f_3 x_3 \dots f_k xk}{f_1 + f_2 + f_3 \dots f_k} = \frac{\sum fx}{\sum f}$$

MEASURE OF CENTRAL TENDENCY

For ungrouped data

$$\overline{X} = \frac{x_1 + x_2 + x_3 + \dots x_n}{n} = \frac{\sum X}{n}$$

$$\overline{X} = \frac{2929}{30} = 97.63$$

MEASURE OF CENTRAL TENDENCY

For grouped data

$$\overline{X} = \frac{f_1 x_1 + f_2 x_2 + f_3 x_3 \ldots f_k xk}{f_1 + f_2 + f_3 \ldots f_k} = \frac{\sum fx}{\sum f}$$

$$\overline{X} = \frac{\sum fx}{\sum f} = \frac{2935}{30} = 97.83$$

MEASURE OF CENTRAL TENDENCY

Class Limits	Mid Points (X)	Frequency (*f*)	*f* X
86---90	88	6	528
91---95	93	4	372
96---100	98	10	980
101---105	103	6	618
106---110	108	3	324
111---115	113	1	113
Total	-	30	2935

MEASURE OF CENTRAL TENDENCY

<u>PROPERTIES OF ARITHMETIC MEAN</u>

(i) The sum of the deviations of the observations from their mean is equal to zero

(ii) The sum of squared deviations of the observations from their mean is minimum.

(iii) If n_1 values have mean X_1, n_2 values have mean X_2, n_3 values have mean X_3 and so on then the mean of all values is.

$$\overline{X} = \frac{n_1 \overline{X}_1 + n_2 \overline{X}_2, \ldots n_k \overline{X}_k}{n_1 + n_2 + \ldots n_k} = \frac{\sum nx}{\sum n}$$

MEASURE OF CENTRAL TENDENCY

PROPERTIES OF ARITHMETIC MEAN

(iv) If a constant 'a' is added (subtracted) to each observation of a variable X, then the mean will increased (decreased) by that constant .

(v) If each observation of the variable X is multiplied (divided) by a constant 'a', then the mean will be multiplied (divided) by that constant.

MEASURE OF CENTRAL TENDENCY

<u>Weighted Arithmetic mean</u>

Sometimes, we want to find the average of certain values which are not of equal importance. When the values are not of equal importance we assign them certain numerical values to express their relative importance. These numerical values are called weights. If X1,X2,...Xk have weights W1, W2, ...Wk, then the weighted mean is defined as

$$\overline{X}_w = \frac{W_1 X_1 + W_2 X_2 + \dots W_{kX_k}}{W_1 + W_2 + \dots W_k} = \frac{\sum WX}{\sum W}$$

MEASURE OF CENTRAL TENDENCY

Weighted Arithmetic mean

The data is about the percentage kill (y) and the no of insects (w) used in study, the interest is to calculate the mean of the percentage kill.

Y	88	85.7	52.1	33.3	12	Sum
W	44	42	24	16	6	132
YW	3872	3599.4	1250.4	532.8	72	9326.6

$$\overline{Y}_w = \frac{88*44+85.7*42+52.1*24+33.3*16+6*12}{44+42+24+16+6} = \frac{9326.6}{132} = 70.65$$

MEASURE OF CENTRAL TENDENCY

Geometric mean

The geometric mean, G of a set of n positive values X1, X2, X3...Xn is the positive nth root of the product of the values. For ungrouped data Geometric mean is given by

$$G = (X_1 \, x \, X_2 \, x \, X_3 \ldots X_n)^{\frac{1}{n}}$$

For calculation use

$$\text{Log}\, G = \frac{[\log x_1 + \log x_2 \ldots \log X_n]}{n} = \frac{\sum \log X}{n}$$

$$G = Antilog\left(\frac{\sum \log X}{n}\right)$$

MEASURE OF CENTRAL TENDENCY

Geometric mean

For grouped data Geometric mean is given by

$$G = [X_1^{f_1} \, x \, X_2^{f_2} \, x \, X_3^{f_3} \ldots X_k^{f_k}]^{\frac{1}{n}}$$

Where k denote total number of classes, and n denote total number of values. For calculation use.

$$\text{Log}\, G = \frac{f_1 \log X_1 + f_2 \log X_2 + f_3 \log X_3 \ldots f_k \log X_k}{f_1 + f_2 + f_3 \ldots f_k}$$

MEASURE OF CENTRAL TENDENCY

Geometric mean

$$\operatorname{Log} G = \frac{\sum f \log X}{\sum f}$$

$$G = Antilog \left(\frac{\sum f \log X}{\sum f} \right)$$

Geometric mean is useful when dealing with relative values such as to find the average of percentage changes, independent ratios and index number.

MEASURE OF CENTRAL TENDENCY

Geometric mean (Ungrouped data)

The data is the percentage changes in the weights of 8 animals.

X	45	30	35	40	44	32	42	37
Log(X)	**1.65**	**1.48**	**1.54**	**1.60**	**1.64**	**1.51**	**1.62**	**1.57**

$$G = Antilog\left(\frac{\sum \log X}{n}\right) = Antilog\left(\frac{12.61}{8}\right)$$

$$= Antilog\,(1.576) = 37.67$$

MEASURE OF CENTRAL TENDENCY

Geometric mean (Grouped Data)

The grouped data is available on insect growth population for age and corresponding frequencies.

CLASS	f	X	log(X)	f log(X)
0--4	2	2	0.3010	0.6021
4--8	5	6	0.7782	3.8908
8--12	7	10	1.0000	7.0000
12--16	8	14	1.1461	9.1690
16--20	7	18	1.2553	8.7869
20--24	4	22	1.3424	5.3697
24--28	1	26	1.4150	1.4150
TOTAL	**34**			**36.2334**

MEASURE OF CENTRAL TENDENCY

<u>Geometric mean (Grouped Data)</u>

$$G = Antilog\left(\frac{\sum f \log X}{\sum f}\right) = Antilog\left(\frac{36.2334}{34}\right)$$

$$= Antilog\ (1.0657) = 11.6329$$

MEASURE OF CENTRAL TENDENCY

Harmonic mean

The Harmonic mean, H of a set of n values

X1, X2, X3,...Xn

is defined as the reciprocal of the arithmetic mean of the reciprocals of the values.

The harmonic mean is particularly useful when dealing with the averages of certain types of rates and ratios.

MEASURE OF CENTRAL TENDENCY

Harmonic mean

For ungrouped data

$$H = Reciprocal\,of\left(\frac{\frac{1}{x_1}+\frac{1}{x_2}+\frac{1}{x_3}+\ldots\frac{1}{x_n}}{n}\right) = \frac{n}{\sum\frac{1}{X}}$$

For grouped data

$$H = Reciprocal\ of\left(\frac{\frac{f_1}{x_1}+\frac{f_2}{x_2}+\frac{f_3}{x_3}+\ldots\frac{f_k}{x_k}}{f_1+f_2+f_3\cdots f_k}\right) = \frac{\sum f}{\sum f\,\frac{1}{x}}$$

MEASURE OF CENTRAL TENDENCY

Harmonic mean

A tractor is running at the rate of 10 Km/hr. during the first 60 Km; at 20 Km/hr. during second 60 Km; 30 Km/hr. during the third 60 Km; 40 Km/hr. during the fourth 60 Km and 50 Km/hr. during the (last) fifth 60 km. what would be the average speed?

$$H = Reciprocal of \left(\frac{\frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} + \ldots \frac{1}{x_n}}{n} \right) = \frac{n}{\sum \frac{1}{X}}$$

$$H = Reciprocal of \left(\frac{\frac{1}{10} + \frac{1}{20} + \frac{1}{30} + \frac{1}{40} + \frac{1}{50}}{5} \right) = \frac{5}{0.2283}$$

$$\mathrm{H} = 21.898\, km/hr$$

MEASURE OF CENTRAL TENDENCY

Harmonic mean (Grouped data)

Example:- Calculate Harmonic mean for the following data

CLASS	f	X	1/X	f (1/X)
0--4	2	2	0.5000	1.0000
4--8	5	6	0.1667	0.8333
8--12	7	10	0.1000	0.7000
12--16	8	14	0.7114	0.5714
16--20	7	18	0.0556	0.3889
20--24	4	22	0.0385	0.1818
24--28	1	26	1.4150	0.0385
TOTAL	**34**			**3.7139**

MEASURE OF CENTRAL TENDENCY

Harmonic mean (Grouped data)

$$H = \frac{\sum f}{\sum \frac{f}{x}} = \frac{34}{3.7139} = 9.15$$

Relation between A.M, G.M AND H.M

A.M ≥ G.M ≥ H.M

The three means are equal only when all the observations are identical.